

Question 1a)

Steel Alloys: $J \in \{A, B\}$, Raw Metals: $I = \{1, 2, 3, 4, 5\}$, Elements: $E = \{Fe, C, Cr, Ni, Mn\}$

Let r_j be the revenue produced from selling one ton of Steel j , $\forall j \in J$.

Let c_i be the cost incurred from buying one ton of raw metal i , $\forall i \in I$.

Let A_i be the availability (supply constraint) of raw metal i , $\forall i \in I$.

Let z_j be the number of tonnes of Steel j produced/sold, $\forall j \in J$.

Let y_i be the number of tonnes of raw metal i purchased, $\forall i \in I$.

Let x_{ij} be the amount of raw metal i used to obtain steel alloy j , $\forall i \in I, \forall j \in J$.

Let p_{ie} be the percentage of element e associated with each raw metal i , $\forall i \in I, e \in E$.

Let l_{ej} be the lower bound of the percent Elemental composition for element $e \in E$ for alloy J .

Let u_{ej} be the upper bound of the percent Elemental composition for element $e \in E$ for alloy J .

$$A = \begin{bmatrix} 500 \\ 50 \\ 3000 \\ 900 \\ 200 \end{bmatrix}, P = \begin{bmatrix} 96 & 3.5 & 0 & 0 & 0.5 \\ 91 & 1.5 & 2 & 1.5 & 4 \\ 99.75 & 0.05 & 0 & 0 & 0.2 \\ 60 & 0.5 & 38 & 0 & 1.5 \\ 55.5 & 0.1 & 0 & 44 & 0.5 \end{bmatrix}, l = \begin{bmatrix} 69 & 0 & 18 & 8 & 0 \\ 90 & 0.05 & 0 & 0 & 0.5 \\ 71 & 0.3 & 20 & 10 & 1 \\ 95 & 0.25 & 3 & 3 & 2 \end{bmatrix}, u = \begin{bmatrix} 69 & 0 & 18 & 8 & 0 \\ 90 & 0.05 & 0 & 0 & 0.5 \\ 71 & 0.3 & 20 & 10 & 1 \\ 95 & 0.25 & 3 & 3 & 2 \end{bmatrix}, r = \begin{bmatrix} 2200 \\ 1200 \end{bmatrix}, C = \begin{bmatrix} 400 \\ 500 \\ 650 \\ 800 \\ 1500 \end{bmatrix}$$

Objective function: $\max \sum_{j \in J} r_j z_j - \sum_{i \in I} c_i y_i$

Constraints: s.t. $y_i \leq A_i \quad \forall i \in I$

$$z_j = \sum_{i \in I} x_{ij} \quad \forall j \in J$$

$$y_i = \sum_{j \in J} x_{ij} \quad \forall i \in I$$

$$l_{ej} z_j \leq \sum_{i \in I} p_{ie} x_{ij} \leq u_{ej} z_j \quad \forall e \in E, \forall j \in J$$

$$x_{ij} \geq 0$$

$$y_i \geq 0$$

$$z_j \geq 0$$

Availability constraint

Alloy balance; mass conservation
for alloy j # Inputs = Outputs

Purchase allocation; Use what
you buy for raw metal i .

Element e in alloy j must
remain within spec.

1b) See code in rough work. optimal solution given by.

[15]: m.printAttr('X') # print variable values in the optimal solution of m

Variable	X
z_A	948.106
z_B	1029.91
y_1	29.8417
y_2	58
y_3	1125.83
y_4	572.345
y_5	200
x_1_A	1.81355
x_3_A	288.241
x_4_A	493.668
x_5_A	172.383
x_1_B	28.0282
x_2_B	50
x_3_B	845.591
x_4_B	78.6774
x_5_B	27.6171

Words: The optimal solution is given by producing 948.106t of Stainless (A) and 1029.91t of HSLA (B). We buy 29.8417t of pig iron, 50t scrap, 1125.83t weight iron, 572.345t ferro-chromium, 200.000t ferromanganese. The stainless blend is made of 1.8135t pig iron, no scrap, 288.241t weight, ferromanganese of 493.668t and ferromanganese of 172.383t. The HSLA (B) is made of 27.617t pig iron, 50t scrap, 845.591t weight, 78.6774t ferro-chromium, and 27.617t ferromanganese. The economic income is revenue - costs. Revenue = $2200 \times 948.106 + 1200 \times 1029.91 = \321725
cost = $400 \times 29.8417 + 500 \times 50 + 650 \times 1125.83 + 800 \times 572.345 + 1500 \times 200 = \1526602
Profit = $\$321725 - \$1526602 = \$1195123$

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[15]: import math
from gurobipy import GRB

m.setParam('Method', 0) # simplex so sensitivity attributes are available
m.optimize()

zA = m.getVarByName('z_A')
lo, hi = m.SABObjLow, m.SABObjUp # allowable range for the coef of z_A

print("Continuous range keeping current basis:", lo, "to", hi)

# Integer p that keep the same optimal solution (inclusive)
p_min = math.ceil(lo)
p_max = math.floor(hi)
print("Integer p range (inclusive):", p_min, "to", p_max)

Set parameter Method to value 0
Gurobi Optimizer version 12.0.3 build v12.0.3rc8 (mac64) - Darwin 25.0.0 (25A354)
CPU model: Apple M1
Thread count: 8 physical cores, 8 logical processors, using up to 8 threads

Non-default parameters:
Method 0

Optimize a model with 28 rows, 17 columns and 118 nonzeros
Coefficient statistics:
  Matrix range [1e+04, 1e+08]
  Objective range [4e+02, 2e+03]
  Bounds range [0e+00, 0e+08]
  RHS range [1e+01, 3e+03]

Solved in 0 iterations and 0.00 seconds (0.00 work units)
Optimal objective 1.79512305e+06
Continuous range keeping current basis: 888.488724538486 to 3811.386851812819
Integer n range (inclusive): 889 to 3811

from gurobipy import GRB
def plot_ranges(m, x, y):
    return tuple((x, y, m.getAttr('x', x), m.getAttr('y', y)))

m.setParam('PlotRange', 0)
m.optimize()

baseline_obj = m.ObjVal
zA = m.getVarByName('z_A')
p_min, p_max = p_min, p_max
interval_zA_obj = (baseline_obj, baseline_obj)

print('obj = %s' % obj)
for p in range(p_min, p_max + 1):
    m.setParam('PlotRange', 0)
    m.optimize()
    obj = m.ObjVal
    interval_zA_obj = (interval_zA_obj, obj)
    print('Baseline obj kept for p (%s): %s' % (p, obj))

if p_min <= p_max:
    interval_zA_obj = (interval_zA_obj, obj)
    print('Baseline obj kept for p (%s): %s' % (p, obj))

else:
    interval_zA_obj = (interval_zA_obj, obj)
    print('Baseline obj kept for p (%s): %s' % (p, obj))

for p in range(p_min, p_max + 1):
    m.setParam('PlotRange', 0)
    m.optimize()
    obj = m.ObjVal
    interval_zA_obj = (interval_zA_obj, obj)
    print('Baseline obj kept for p (%s): %s' % (p, obj))

for p in range(p_min, p_max + 1):
    m.setParam('PlotRange', 0)
    m.optimize()
    obj = m.ObjVal
    interval_zA_obj = (interval_zA_obj, obj)
    print('Baseline obj kept for p (%s): %s' % (p, obj))
```

1c) I used Gurobi's Sensitivity attributes (SABObjLow/SABObjUp) for variable z_A (stainless steel) which is the objective coeff (price) of stainless steel. The interval [SABObjLow/SABObjUp] is the range of prices for which the current optimal basis remains optimal. Also checked with sweeping integer prices as a backup.

Answer: [889 - 3911]

Question 2:

Let P be the house price at the start of year 1.

Let S be the starting cash at the start of year 1 (given).

Let T_i represent the income received at the end of year i , $\forall i \in \{1, \dots, 20\}$

Let L_A represent the amount of Mortgage A borrowed at the start of Year 1.

Let L_{B1} represent the amount of the first mortgage B borrowed at the start of Year 1.

Let L_{B2} represent the amount of the first mortgage B borrowed at the start of Year 11.

Let M_A^i represent the remaining principal of mortgage A at the start of Year i , for $i=1, \dots, 21$

Let M_{B1}^i represent the remaining principal of first mortgage B at the start of Year i , for $i=1, \dots, 11$

Let M_{B2}^i represent the remaining principal of second mortgage B at the start of Year i , for $i=11, \dots, 21$

Let P_A^i represent the amount of Mortgage A principal paid at the end of year i , for $i=1, \dots, 20$

Let P_{B1}^i represent the amount of first mortgage B principal paid at the end of year i , for $i=1, \dots, 10$

Let P_{B2}^i represent the amount of second mortgage B principal paid at the end of year i , for $i=11, \dots, 20$

Let C_i represent the cash available at the start of year i , for $i=1, \dots, 21$

$$P = 1000000$$

$$\text{Objective Function: } \min \sum_{i=1}^{20} 0.10 M_A^i + \sum_{i=1}^{10} 0.07 M_{B1}^i + \sum_{i=11}^{20} 0.07 M_{B2}^i$$

Constraints:

$$\text{s.t. } S + L_A + L_{B1} = P + C_1$$

$$M_A^{i+1} = M_A^i - P_A^i \quad (i=1, \dots, 20)$$

$$M_A^1 = L_A, \quad M_A^{21} = 0$$

$$0.03 L_A \leq P_A^i \leq 0.07 L_A \quad (i=1, \dots, 20)$$

$$M_{B1}^{i+1} = M_{B1}^i - P_{B1}^i \quad (i=1, \dots, 10)$$

$$M_{B1}^1 = L_{B1}, \quad M_{B1}^{11} = 0$$

$$0.05 L_{B1} \leq P_{B1}^i \leq 0.15 L_{B1} \quad (i=1, \dots, 10)$$

$$M_{B2}^{i+1} = M_{B2}^i - P_{B2}^i \quad (i=11, \dots, 20)$$

$$M_{B2}^{11} = L_{B2}, \quad M_{B2}^{21} = 0$$

$$0.05 L_{B2} \leq P_{B2}^i \leq 0.15 L_{B2} \quad (i=11, \dots, 20)$$

$$C_{i+1} = 1.05 C_i + T_i - (P_A^i + 0.10 M_A^i) - (P_{B1}^i + 0.07 M_{B1}^i) \quad (i=1, \dots, 9)$$

$$C_{11} = 1.05 C_{10} + T_{10} - (P_A^{10} + 0.10 M_A^{10}) - (P_{B1}^{10} + 0.07 M_{B1}^{10}) + L_{B2}$$

$$C_{i+1} = 1.05 C_i + T_i - (P_A^i + 0.10 M_A^i) - (P_{B2}^i + 0.07 M_{B2}^i) \quad (i=11, \dots, 20)$$

$$S, L_A, L_{B1}, L_{B2}, M_A^i, M_{B1}^i, M_{B2}^i, P_A^i, P_{B1}^i, P_{B2}^i, C_i, T_i \geq 0$$

Question 3:

a) Let x = Kilograms of yarn purchased

Let $f_i(x)$ = cost functions for yarn, $i \in I$.

$$I = \{1, 2, 3\}$$

Price Functions:

$$f_1(x) = 50x, \quad f_2(x) = 65x - 1500, \quad f_3(x) = 75x - 4000$$

Derive total cost as:

$$C = \max\{f_1(x), f_2(x), f_3(x)\}, \quad i \in I$$

Objective Function: $\min C$

constraints: s.t. $C \geq 50x$

$$C \geq 65x - 1500$$

$$C \geq 75x - 4000$$

→ The supplier's pricing scheme is piecewise linear and non-decreasing. The actual cost function is the "upper envelope" of the three lines. By introducing the auxiliary variable C and forcing it to dominate each linear segment, minimize C correctly captures the convex piecewise-linear cost of yarn.

b) Alice's actions: $A = \{1, 3, 5, 7, 9\}$, Bob's actions: $B = \{2, 4, 6, 8\}$

Let p_i be the probability Alice plays the i th action (so p_i is prob of 1, p_2 of 3, etc...)

Alice's guaranteed value V : the minimum expected payoff she can secure against any (best-response) choice of Bob.

Payoff matrix:

(Alice's perspective):

	2	4	6	8
1	2	-1	-1	-1
3	2	4	-3	-3
5	5	4	6	-5
7	-7	-7	6	8
9	-9	-9	-9	8

ex. for us $i=2$; expected payoff

$$\text{is } 2p_1 + 2p_2 - 5p_3 - 7p_4 - 9p_5.$$

Alice's LP (maxmin):

max V

$$\text{s.t. } 2p_1 + 2p_2 - 5p_3 - 7p_4 - 9p_5 \geq V \quad (\text{Bob plays 2})$$

$$1p_1 + 4p_2 + 4p_3 - 7p_4 - 9p_5 \geq V \quad (\text{Bob plays 4})$$

$$1p_1 - 3p_2 + 6p_3 + 6p_4 - 9p_5 \geq V \quad (\text{Bob plays 6})$$

$$1p_1 - 3p_2 - 5p_3 + 8p_4 + 8p_5 \geq V \quad (\text{Bob plays 8})$$

$$p_1 + p_2 + p_3 + p_4 + p_5 = 1, \quad p_i \geq 0 \quad (i=1, \dots, 5)$$

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[12]: m.printAttr('X') # print var
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Variable	X
z	0.0925926
p1	0.37037
p2	0.308889
p3	0.037037
p4	0.203704

c) In Wloo, Timmies has long lines across campus, especially at peak hours in SLC, DC, and occasionally ML. Suppose Food Services wants to assign extra staff hours to each location to help speed up service. The goal is to minimize the longest expected wait times across all Tim Hortons on campus.

Sample problem:

Let L = set of all Tim Hortons on campus

Let d_L = average demand (students/hour) at location $L \in L$

Let h_L = staff hours allocated to location L (decision variables)

Let $w_L = \frac{d_L}{h_L}$ = estimated average wait time at location L .

Let E = maximum wait time across all locations.

Formulation: $\min E$

$$\text{s.t. } E \geq w_L = \frac{d_L}{h_L} \quad \forall L \in L$$

$$\sum_{L \in L} h_L \leq H \quad (\text{total staff hours available})$$

$$h_L \geq 0 \quad \forall L \in L$$

Explanation: By minimizing E , Waterloo ensures that no single Tim Hortons line on campus gets unbearably long. This balances wait times across student hotspots and directly improves daily student life.